

Retail Sales Analytics Dashboard

Project Overview

This project demonstrates an interactive Excel dashboard built using Pivot Tables, Pivot Charts, Slicers and a Timeline based on a retail sales dataset.

Dataset Source

Retail Sales Dataset from Kaggle

<https://www.kaggle.com/datasets/mohammadtalib786/retail-sales-dataset>

Workflow

- Download dataset from Kaggle

Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total	Day	Day_Sort	Month	Month_Sort	Age_Group	Quantity_Category	Weekday_flag
1	2023/11/24	CUST001	Male	34	Beauty	3	50	150	Friday		5	November			
2	2023/02/27	CUST002	Female	26	Clothing	2	500	1000	Monday		1	February			
3	2023/01/13	CUST003	Male	50	Electronics	1	30	30	Friday		5	January			
4	2023/05/21	CUST004	Male	37	Clothing	1	500	500	Sunday		7	May			
5	2023/05/06	CUST005	Male	30	Beauty	2	50	100	Saturday		6	May			
6	2023/04/25	CUST006	Female	45	Beauty	1	30	30	Tuesday		2	April			
7	2023/03/13	CUST007	Male	46	Clothing	2	25	50	Monday		1	March			
8	2023/02/22	CUST008	Male	30	Electronics	4	25	100	Wednesday		3	February			
9	2023/12/13	CUST009	Male	63	Electronics	2	300	600	Wednesday		3	December			
10	2023/10/07	CUST010	Female	52	Clothing	4	50	200	Saturday		6	October			
11	2023/02/14	CUST011	Male	23	Clothing	2	50	100	Tuesday		2	February			
12	2023/10/30	CUST012	Male	35	Beauty	3	25	75	Monday		1	October			
13	2023/08/05	CUST013	Male	22	Electronics	3	500	1500	Saturday		6	August			
14	2023/01/17	CUST014	Male	64	Clothing	4	30	120	Tuesday		2	January			
15	2023/01/16	CUST015	Female	42	Electronics	4	500	2000	Monday		1	January			
16	2023/02/17	CUST016	Male	19	Clothing	3	500	1500	Friday		5	February			
17	2023/04/22	CUST017	Female	27	Clothing	4	25	100	Saturday		6	April			
18	2023/04/30	CUST018	Female	47	Electronics	2	25	50	Sunday		7	April			
19	2023/09/16	CUST019	Female	67	Clothing	2	25	50	Saturday		6	September			

- To separate each field into columns,
 - Select Column A.
 - Go to Data → Text to Columns.
 - Choose Delimited.
 - Click Next.
 - Select Semicolon.
 - Click Finish.

Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount	Day	Day_Sort	Month	Month_Sort	Age_Group	Quantity_Category	Weekday_flag
1	24-11-2023	CUST001	Male	34	Beauty	3	50	150	Friday		5	November			
2	27-02-2023	CUST002	Female	26	Clothing	2	500	1000	Monday		1	February			
3	13-01-2023	CUST003	Male	50	Electronics	1	30	30	Friday		5	January			
4	21-05-2023	CUST004	Male	37	Clothing	1	500	500	Sunday		7	May			
5	06-05-2023	CUST005	Male	30	Beauty	2	50	100	Saturday		6	May			
6	25-04-2023	CUST006	Female	45	Beauty	1	30	30	Tuesday		2	April			
7	13-03-2023	CUST007	Male	46	Clothing	2	25	50	Monday		1	March			
8	22-02-2023	CUST008	Male	30	Electronics	4	25	100	Wednesday		3	February			
9	13-12-2023	CUST009	Male	63	Electronics	2	300	600	Wednesday		3	December			
10	07-10-2023	CUST010	Female	52	Clothing	4	50	200	Saturday		6	October			
11	14-02-2023	CUST011	Male	23	Clothing	2	50	100	Tuesday		2	February			
12	30-10-2023	CUST012	Male	35	Beauty	3	25	75	Monday		1	October			
13	05-08-2023	CUST013	Male	22	Electronics	3	500	1500	Saturday		6	August			
14	17-01-2023	CUST014	Male	64	Clothing	4	30	120	Tuesday		2	January			
15	16-01-2023	CUST015	Female	42	Electronics	4	500	2000	Monday		1	January			
16	17-02-2023	CUST016	Male	19	Clothing	3	500	1500	Friday		5	February			
17	22-04-2023	CUST017	Female	27	Clothing	4	25	100	Saturday		6	April			
18	30-04-2023	CUST018	Female	47	Electronics	2	25	50	Sunday		7	April			
19	16-09-2023	CUST019	Female	67	Clothing	2	25	50	Saturday		6	September			

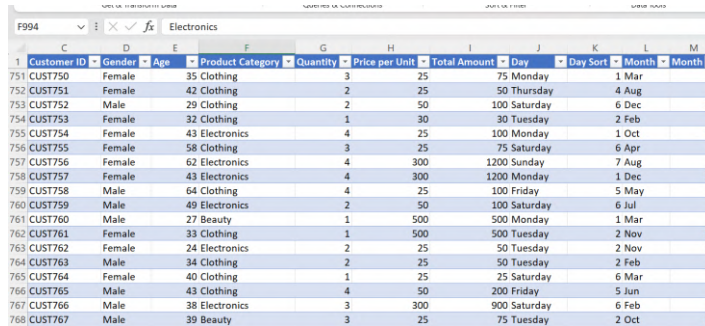
- Save the File as an Excel Workbook

If your dataset is a CSV:

1. Click File → Save As.
2. Choose a location.
3. Set Save as type to Excel Workbook (*.xlsx).
4. Save it with a name like:

Retail Sales Dashboard.xlsx

- Convert data into an Excel Table using (Ctrl+T).



	C	D	E	F	G	H	I	J	K	L	M
	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount	Day	Day Sort	Month	Month Sort
751	CUST750	Female	35	Clothing	3	25	75	Monday		1 Mar	3
752	CUST751	Female	42	Clothing	2	25	50	Thursday		4 Aug	8
753	CUST752	Male	29	Clothing	2	50	100	Saturday		6 Dec	12
754	CUST753	Female	32	Clothing	1	30	30	Tuesday		2 Feb	2
755	CUST754	Female	43	Electronics	4	25	100	Monday		1 Oct	10
756	CUST755	Female	58	Clothing	3	25	75	Saturday		6 Apr	4
757	CUST756	Female	62	Electronics	4	300	1200	Sunday		7 Aug	8
758	CUST757	Female	43	Electronics	4	300	1200	Monday		1 Dec	12
759	CUST758	Male	64	Clothing	4	25	100	Friday		5 May	5
760	CUST759	Male	49	Electronics	2	50	100	Saturday		6 Jul	7
761	CUST760	Male	27	Beauty	1	500	500	Monday		1 Mar	3
762	CUST761	Female	33	Clothing	1	500	500	Tuesday		2 Nov	11
763	CUST762	Female	24	Electronics	2	25	50	Tuesday		2 Nov	11
764	CUST763	Male	34	Clothing	2	25	50	Tuesday		2 Feb	2
765	CUST764	Female	40	Clothing	1	25	25	Saturday		6 Mar	3
766	CUST765	Male	43	Clothing	4	50	200	Friday		5 Jun	6
767	CUST766	Male	38	Electronics	3	300	900	Saturday		6 Feb	2
768	CUST767	Male	39	Beauty	3	25	75	Tuesday		2 Oct	10

- **Freeze the Header Row**

This is especially useful for large datasets.

Go to:

View → Freeze Panes → Freeze Top Row

Now, when you scroll, the column names remain visible.

- **Clean the data (check blanks, duplicates, formats).**

Check for Duplicate Records

1. Select the entire dataset.
2. Go to:
 - o **Data → Remove Duplicates**
 - o A dialog box appears.
 - o Select **all columns**.
 - o Click **OK**.

Check for Missing Values

Blank cells are one of the biggest data quality issues.

- o Press **Ctrl + G**
 - o Choose **Special**
 - o Select **Blanks**
 - o Click **OK**
- Excel highlights every blank cell.

- **Validate Data Types**

Every column should contain the correct type of data.

Column	Correct Type
Transaction ID	Text/Number
Date	Date
Gender	Text
Age	Number
Quantity	Whole Number
Price per Unit	Currency
Total Amount	Currency

- **Check Date**

If the date is left aligned, it may be stored as text.

Convert it if necessary.

- Create helper columns: Month, Month_Sort, Day, Day_Sort, Age_Group, Weekday_Flag.

Helper Column	Formula	Purpose
Month	=TEXT([@Date],"mmm")	Monthly analysis
Month_Sort	=MONTH([@Date])	Correct month sorting
Day	=TEXT([@Date],"dddd")	Daily analysis
Day_Sort	=WEEKDAY([@Date],2)	Correct weekday sorting
Age_Group	Nested IF	Customer segmentation

Helper Column	Formula	Purpose
Weekday_Flag	=IF(WEEKDAY([@Date],2)<=5,"Weekday","Weekend")	Weekday vs Weekend

- **Steps to Create a PivotTable**

1. **Select the data**

- o Click any cell within your dataset (or select the entire dataset).

2. **Insert a PivotTable**

- o Go to the **Insert** tab.
- o Click **PivotTable**.

3. **Choose the data source**

- o Verify the Table/Range.
- o Select **New Worksheet** (recommended) or **Existing Worksheet**.
- o Click **OK**.

4. **Build the PivotTable**

- o The **PivotTable Fields** pane will appear.
- o Drag fields into the following areas:
 - **Rows** → Displays data vertically.
 - **Columns** → Displays data horizontally.
 - **Values** → Performs calculations (Sum, Count, Average, etc.).
 - **Filters** → Filters the entire PivotTable.

- Create Pivot Tables for KPIs.

Pivot tables were created for each KPIs by dragging the following fields into the **values** area in the field list.

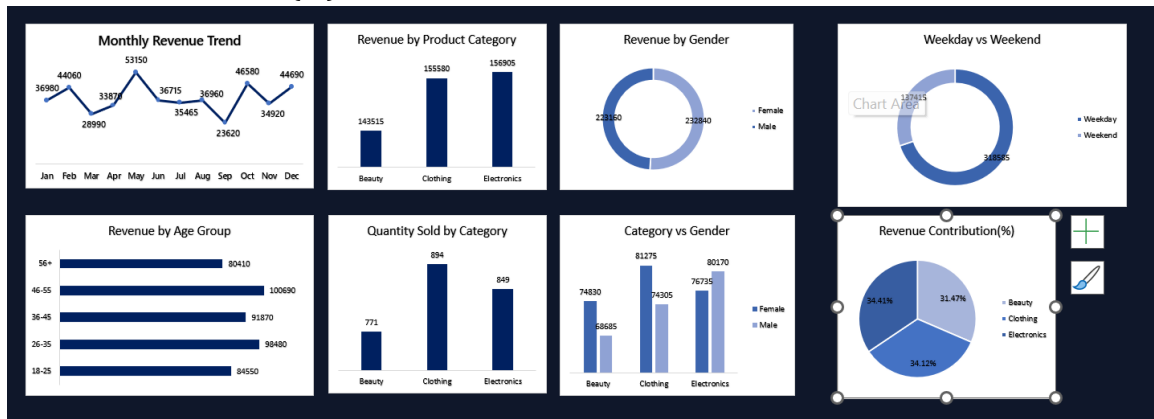
KPI	Values
Total Revenue	Sum of Total Amount
Total Orders	Count of Transaction ID
Quantity Sold	Sum of Quantity
Customers	Count of Customer ID

Charts

- Create Pivot Charts.
 - Click anywhere inside the PivotTable.
 - Go to the **PivotTable Analyze** tab.
 - Click **PivotChart**.
 - Choose a suitable chart type (Column, Bar, Line, Pie, Doughnut, etc.).
 - Click **OK**.
- **Format the PivotChart**
 - Add a meaningful **Chart Title**.
 - Format numbers (Currency, Percentage, Decimal, etc.).
 - Apply a chart style and colours.
 - Remove unnecessary gridlines or legends if required.

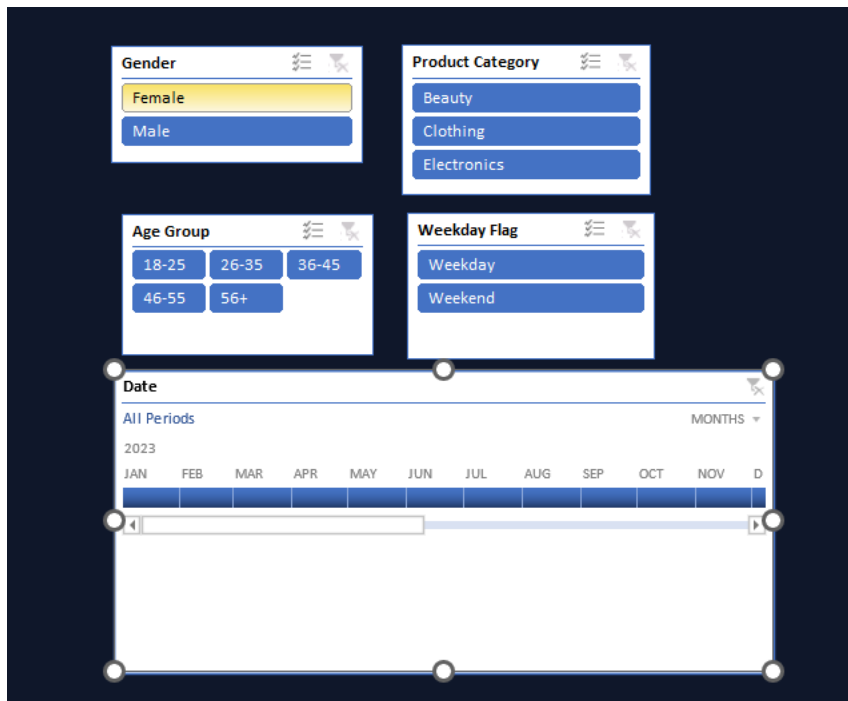
The following charts were created using pivot table

- Monthly Revenue Trend
- Revenue by Product Category
- Revenue by Genders
- Revenue by Age Group
- Quantity Sold by Category
- Category vs Gender
- Weekday vs Weekend
- Revenue Contribution (%)



- Insert Slicers (Gender, Product Category, Age Group, Weekday Flag).
- **Steps to Create a Slicer**
 1. Click anywhere inside the **PivotTable**.
 2. Go to the **PivotTable Analyze** tab.
 3. Click **Insert Slicer**.
 4. Select the fields you want to use as filters (e.g., **Gender, Product Category, Age Group, Weekday Flag**).
 5. Click **OK**.
 6. Move and resize the slicers on the dashboard as required.

7. To connect a slicer to multiple PivotTables:
 - o Click the slicer.
 - o Go to **Slicer → Report Connections** (or **PivotTable Connections**).
 - o Select all the PivotTables you want the slicer to control.
 - o Click **OK**.
- Insert Timeline using Date.
Steps to Create a Timeline
 1. Click anywhere inside the **PivotTable**.
 2. Go to the **PivotTable Analyze** tab.
 3. Click **Insert Timeline**.
 4. Select the **Date** field.
 5. Click **OK**.
 6. Place the timeline on the dashboard.
 7. Use the drop-down menu on the timeline to choose the time level (**Years, Quarters, Months, or Days**).
 8. Drag the timeline slider to filter the PivotTables for a specific time period.
 9. To connect the timeline to multiple PivotTables:
 - o Click the timeline.
 - o Go to **Timeline → Report Connections**.
 - o Select all the PivotTables you want the timeline to control.
 - o Click **OK**.



- Format dashboard with dark theme.

